

OBARSIA LOTRULUI, Romania

WMO: 152970

Lat: 45.4356N

Lon: 23.6308E

Elev: 4429

StdP: 12.49

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.0	(c) 2.1	(d) -9.7	(e) 3.8	(f) -0.7	(g) -4.2	(h) 5.2	(i) 2.7	(j) 11.9	(k) 23.5	(l) 9.8	(m) 21.1	(n) 3.0	(o) 320	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.4	(c) 75.2	(d) 59.4	(e) 72.2	(f) 58.6	(g) 69.5	(h) 57.0	(i) 61.1	(j) 71.6	(k) 59.7	(l) 69.8	(m) 58.4	(n) 67.9	(o) 3.0	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.4	82.9	64.6	55.8	78.2	63.3	54.4	74.1	62.4	29.4	72.1	28.3	70.0	27.4	68.3	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.2	(b) 7.6	(c) 6.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 9.2	(b) 7.6	(c) 6.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	21.6	26.3	31.0	37.5	47.1	54.0	55.9	56.7	48.6	41.8	33.3	25.3	
	DBStd	13.68	9.08	7.57	6.88	7.92	4.78	5.24	4.68	4.24	6.13	5.66	7.07	7.45	
	HDD50	4282	880	663	590	376	113	21	9	6	100	256	501	767	
	HDD65	9133	1345	1083	1055	825	556	331	283	259	493	720	951	1232	
	CDD50	626	0	0	0	2	22	140	192	212	57	1	0	0	
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH74	148	0	0	0	0	3	27	54	63	1	0	0	0	
	CDH80	11	0	0	0	0	1	0	7	3	0	0	0	0	
Wind	WSAvg	2.4	2.8	2.8	2.4	2.3	2.1	2.0	2.1	2.0	2.0	2.6	1.9	3.3	
Precipitation	PrecAvg	32.1	1.5	1.3	1.8	2.6	3.7	4.5	4.6	3.1	3.0	2.3	2.0	2.0	
	PrecMax	39.9	3.1	3.6	3.9	4.4	7.6	8.7	10.4	5.3	5.9	4.6	7.4	4.3	
	PrecMin	21.0	0.4	0.3	0.4	0.7	1.5	1.1	1.7	0.8	0.4	0.0	0.0	0.3	
	PrecStd	5.6	0.8	0.8	0.9	1.0	1.5	1.7	2.2	1.2	1.5	1.3	1.5	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.2	48.6	56.3	65.9	70.6	77.0	79.2	79.2	72.0	67.6	54.5	41.2	
		MCWB	36.6	41.5	44.2	49.0	54.7	59.1	60.0	60.8	55.6	49.6	45.2	35.2	
	2%	DB	38.5	44.0	50.7	61.9	65.8	73.6	74.5	75.1	67.5	63.1	50.2	37.8	
		MCWB	34.7	36.4	40.1	48.1	52.7	58.6	59.9	59.2	54.9	49.0	43.9	33.9	
	5%	DB	35.8	40.7	45.5	57.4	61.8	70.0	71.5	71.6	64.4	58.5	47.0	35.8	
		MCWB	32.8	35.8	38.1	45.7	51.3	57.4	58.9	58.6	53.8	48.0	42.5	33.0	
	10%	DB	33.3	37.6	41.2	51.8	58.2	66.6	68.4	68.5	60.6	53.0	43.8	34.2	
		MCWB	30.8	34.0	35.6	43.6	49.7	55.8	57.2	57.6	53.1	45.7	40.4	32.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	42.0	44.9	51.1	56.1	62.6	62.8	63.4	58.9	52.2	46.7	36.9
			MCDB	40.8	47.5	55.3	63.2	67.3	71.8	72.4	72.9	68.3	61.4	52.9	39.8
2%		WB	35.2	38.2	41.0	49.0	54.0	59.7	60.9	61.3	56.7	50.3	44.9	34.7	
		MCDB	38.1	42.2	48.9	61.0	63.2	71.2	71.4	71.8	64.5	59.8	49.4	36.8	
5%		WB	33.1	36.2	38.5	46.8	52.2	58.1	59.4	59.8	55.1	48.7	43.2	33.5	
		MCDB	35.3	39.8	44.6	56.6	60.6	69.1	69.1	69.7	62.7	57.7	46.7	35.4	
10%		WB	31.3	34.0	36.0	43.8	50.3	56.4	58.2	58.4	53.4	46.2	41.1	32.4	
		MCDB	33.2	37.5	40.4	52.0	57.4	65.7	66.9	67.5	60.0	52.9	44.0	34.4	
Mean Daily Temperature Range		5% DB	MDBR	17.0	19.0	17.0	20.2	19.0	20.9	21.4	22.7	18.9	19.8	15.0	13.5
			MCDBR	18.3	23.7	24.8	30.4	27.3	26.5	28.0	29.3	27.1	32.0	20.9	14.1
	MCWBR		14.7	17.1	16.5	18.1	15.9	14.1	15.3	16.1	15.7	19.8	14.7	11.6	
	5% WB	MCDBR	16.0	19.0	22.5	29.7	25.2	25.3	25.1	25.6	24.4	29.2	18.2	12.3	
		MCWBR	13.3	14.6	15.1	17.9	15.4	14.2	14.3	14.8	15.0	18.4	13.2	10.4	
Clear-Sky Solar Irradiance	taub	0.273	0.300	0.337	0.390	0.376	0.374	0.388	0.401	0.362	0.329	0.304	0.276		
	taud	2.381	2.314	2.299	2.212	2.306	2.338	2.296	2.258	2.353	2.451	2.464	2.423		
	Ebn at Noon	268	279	283	275	283	283	277	269	271	266	253	254		
	Edh at Noon	23	30	36	43	40	39	40	40	33	26	22	20		
All-Sky Solar Radiation	RadAvg	424	677	1027	1378	1643	1840	1823	1655	1184	827	488	348		
	RadStd	52	87	138	153	151	175	152	159	121	97	59	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (81 neighbors)	+1.15	N/A	N/A	+1.62	N/A	N/A	-211	-315	+167	+76

Nomenclature: See separate page